The Biconvex Relaxation of Gromov–Wasserstein Transport

A measure-theoretic account of tightness and alternating minimization

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Abstract

A standard Gromov–Wasserstein algorithm freezes one occurrence of the coupling in the quadratic distortion and optimizes the other. This note presents the underlying mechanism directly on a convex family of measures $\mathcal{K} \subset \mathcal{P}(\mathcal{Z})$. A symmetric bilinear form $\mathcal{E}$ that is nonpositive on feasible differences yields a tight two-measure relaxation: replacing $\mathcal{E}(\pi, \pi)$ by $\mathcal{E}(\pi, \xi)$ changes neither the minimum value nor its diagonal minimizers. Squared Gromov–Wasserstein transport is the running example, with $\mathcal{Z} = \mathcal{X} \times \mathcal{Y}$ and $\mathcal{K} = \Pi(\alpha, \beta)$; negative-type geometry supplies the required sign. The same framework includes both unregularized GW and its entropic regularization. Exact alternating minimization decreases the diagonal objective. For entropic GW on compact spaces, its decrease is exactly the sum of a relative entropy and the nonnegative curvature defect of the GW form. The iterates remain in a uniformly positive, equicontinuous set of coupling densities. This entropy geometry gives sufficient decrease and a projected-gradient error bound without introducing a coordinate Frobenius norm or the much stronger total-variation topology. An analytic Fredholm Łojasiewicz–Simon argument then proves finite length and convergence of the complete measure-valued sequence to a stationary coupling. An abstract version of the argument applies to more general analytic biconvex energies.

1 A quadratic problem over measures

The biconvex mechanism is most transparent before choosing coordinates or discretizing the underlying spaces. Let $\mathcal{Z}$ be a compact metric space, let $\mathcal{M}(\mathcal{Z})$ be the vector space of finite signed Borel measures, and let $\mathcal{K} \subset \mathcal{P}(\mathcal{Z})$ be a nonempty narrowly compact convex set. The two variables of the relaxation will be denoted $\pi, \xi \in \mathcal{K}$.

Let $\mathcal{E} : \mathcal{M}(\mathcal{Z}) \times \mathcal{M}(\mathcal{Z}) \to \mathbb{R}$ be a symmetric bilinear form that is continuous on $\mathcal{K} \times \mathcal{K}$. The main example is generated by a bounded continuous symmetric kernel $L : \mathcal{Z}^2 \to \mathbb{R}$:

$$\mathcal{E}(\pi, \xi) = \iint_{\mathcal{Z}^2} L(z, z') \, \mathrm{d}\pi(z) \, \mathrm{d}\xi(z'). \tag{1}$$

Freezing ξ produces the linear cost

$$\mathcal{C}_\xi(z) := \int_{\mathcal{Z}} L(z, z') \, \mathrm{d}\xi(z'), \qquad \mathcal{E}(\pi, \xi) = \int_{\mathcal{Z}} \mathcal{C}_\xi \, \mathrm{d}\pi. \tag{2}$$

Definition 1.1 (Negative form on feasible differences)**.** The bilinear form $\mathcal{E}$ is *negative on feasible differences* when

$$\mathcal{E}(\sigma, \sigma) \leq 0 \qquad \text{for every } \sigma \in \operatorname{span}(\mathcal{K} - \mathcal{K}). \tag{3}$$

It is strictly negative when equality is possible only for $\sigma = 0$.

This is conditional rather than global negativity: differences of probability measures have zero total mass, and differences of transport plans have zero marginals. These linear conservation laws are precisely what makes the GW form negative.

The running GW example. Let $\mathbb{X} = (\mathcal{X}, d_{\mathcal{X}}, \alpha)$ and $\mathbb{Y} = (\mathcal{Y}, d_{\mathcal{Y}}, \beta)$ be compact metric-measure spaces. Set

$$\mathcal{Z} = \mathcal{X} \times \mathcal{Y}, \qquad \mathcal{K} = \Pi(\alpha, \beta), \qquad L((x,y),(x',y')) = \big(d_{\mathcal{X}}(x,x') - d_{\mathcal{Y}}(y,y')\big)^2. \tag{4}$$

Then

$$\mathrm{GW}_2^2(\mathbb{X}, \mathbb{Y}) = \min_{\pi \in \Pi(\alpha,\beta)} \mathcal{E}(\pi, \pi). \tag{5}$$

For discrete measures $\alpha = \sum_i a_i \delta_{x_i}$ and $\beta = \sum_j b_j \delta_{y_j}$, a coupling is represented by $\mathrm{P} \in \mathrm{U}(\mathrm{a}, \mathrm{b})$ and (1) becomes

$$\mathcal{E}(\mathrm{P}, \mathrm{Q}) = \sum_{i,i',j,j'} \big(D_{ii'} - D'_{jj'}\big)^2 P_{ij} Q_{i'j'}. \tag{6}$$

Here $\mathrm{a} = (a_i)_i$, $\mathrm{b} = (b_j)_j$, $D_{ii'} = d_{\mathcal{X}}(x_i, x_{i'})$, and $D'_{jj'} = d_{\mathcal{Y}}(y_j, y_{j'})$. The measure notation and matrix notation thus describe the same object.

2 Tightness of the biconvex relaxation

Duplicating the measure always lowers the minimum because diagonal pairs remain feasible. Conditional negativity is exactly what prevents the lower bound from becoming strict. The result is the measure-level form of the quadratic principle of Konno (1976), used for generalized GW by Séjourné et al. (2021).

Theorem 2.1 (Tight biconvex relaxation over measures)**.** *Let $\mathcal{R} : \mathcal{K} \to \mathbb{R} \cup \{+\infty\}$ be proper and narrowly lower semicontinuous, and define*

$$\mathcal{F}(\pi) = \mathcal{E}(\pi, \pi) + \mathcal{R}(\pi), \tag{7}$$

$$\mathcal{B}(\pi, \xi) = \mathcal{E}(\pi, \xi) + \frac{1}{2}\mathcal{R}(\pi) + \frac{1}{2}\mathcal{R}(\xi). \tag{8}$$

Assume (3)*. Then both minima are attained and*

$$\min_{\pi, \xi \in \mathcal{K}} \mathcal{B}(\pi, \xi) = \min_{\pi \in \mathcal{K}} \mathcal{F}(\pi). \tag{9}$$

If (π_*, ξ_*) minimizes $\mathcal{B}$, then

$$\mathcal{F}(\pi_*) = \mathcal{F}(\xi_*) = \min_{\mathcal{K}} \mathcal{F}, \qquad \mathcal{E}(\pi_* - \xi_*, \pi_* - \xi_*) = 0. \tag{10}$$

Every relaxed minimizer is diagonal if $\mathcal{E}$ is strictly negative on nonzero feasible differences or if $\mathcal{R}$ is strictly convex on its finite domain. If $\mathcal{R}$ is convex, $\mathcal{B}$ is convex in each measure separately.

Proof. Compactness and lower semicontinuity give existence. For every $\pi, \xi \in \mathcal{K}$, polarization gives

$$\mathcal{B}(\pi, \xi) = \frac{1}{2}\mathcal{F}(\pi) + \frac{1}{2}\mathcal{F}(\xi) - \frac{1}{2}\mathcal{E}(\pi - \xi, \pi - \xi). \tag{11}$$

The sign condition implies that the right-hand side is at least $[\mathcal{F}(\pi) + \mathcal{F}(\xi)]/2$, hence at least $\min_{\mathcal{K}} \mathcal{F}$. Conversely, $\mathcal{B}(\pi, \pi) = \mathcal{F}(\pi)$, which proves (9). Equality in the two inequalities gives (10).

Strict negativity immediately forces $\pi_* = \xi_*$. If instead $\mathcal{R}$ is strictly convex and $\pi_* \neq \xi_*$, then (10) makes the quadratic part affine on the segment joining them, while $\mathcal{R}$ is strictly below its chord. The midpoint would have a smaller value of $\mathcal{F}$, a contradiction. Separate convexity follows because the bilinear term is linear in either measure. □

Example 2.2 (Why negativity is necessary)**.** Take $\mathcal{Z} = \{-1, 1\}$, identify a signed measure with its first moment $x \in [-1, 1]$, and set $\mathcal{E}(x, y) = xy$, $\mathcal{R} = 0$. The diagonal minimum of x^2 is zero, whereas the two-variable minimum of xy is -1. Duplicating a variable is not a tight relaxation for a positive quadratic form.

Concavity and tangent majorization. The same sign condition has an algorithmic consequence. The diagonal quadratic map is concave along feasible segments and, for all $\pi, \xi \in \mathcal{K}$,

$$\mathcal{E}(\xi, \xi) \leq 2\mathcal{E}(\pi, \xi) - \mathcal{E}(\pi, \pi). \tag{12}$$

Indeed, this is exactly the inequality $\mathcal{E}(\xi - \pi, \xi - \pi) \leq 0$ written after expansion. Thus freezing one occurrence of the measure gives a tangent upper bound, not merely one block of a biconvex objective.

3 Why squared GW has the required sign

The abstract condition becomes geometric for squared GW. Recall that a symmetric kernel d on $\mathcal{X}$ is of negative type when

$$\iint d(x, x')\, \mathrm{d}\eta(x)\, \mathrm{d}\eta(x') \leq 0 \qquad \text{whenever } \eta(\mathcal{X}) = 0. \tag{13}$$

By Schoenberg's theorem, this is equivalent, after the usual harmless normalization on the diagonal, to the existence of a Hilbert space $\mathcal{H}$ and a map $\varphi : \mathcal{X} \to \mathcal{H}$ such that $d(x, x') = \|\varphi(x) - \varphi(x')\|^2$ (Schoenberg, 1938). Euclidean powers $d(x, x') = \|x - x'\|^r$, $0 < r \leq 2$, and tree distances are standard examples.

Proposition 3.1 (Conditional negativity of the squared GW form)**.** *Assume that $d_{\mathcal{X}}$ and $d_{\mathcal{Y}}$ are kernels of negative type. For every signed measure σ on $\mathcal{X} \times \mathcal{Y}$ with zero marginals,*

$$\mathcal{E}(\sigma, \sigma) = -8 \left\| \int \varphi(x) \otimes \psi(y)\, \mathrm{d}\sigma(x, y) \right\|^2_{\mathcal{H}_{\mathcal{X}} \otimes \mathcal{H}_{\mathcal{Y}}} \leq 0, \tag{14}$$

where φ and ψ are Hilbert embeddings of the two kernels. Consequently, the GW specialization (4) *satisfies Definition 1.1.*

Proof. Expand the squared difference in (4). The terms involving only $d_{\mathcal{X}}^2$ or only $d_{\mathcal{Y}}^2$ vanish because σ has zero marginals. Hence

$$\mathcal{E}(\sigma, \sigma) = -2 \iint d_{\mathcal{X}}(x, x') d_{\mathcal{Y}}(y, y')\, \mathrm{d}\sigma(x, y)\, \mathrm{d}\sigma(x', y'). \tag{15}$$

Insert the two squared-Hilbert-distance representations. Every term containing a function of x alone or y alone again vanishes by the zero-marginal constraints. The only surviving term is four times the squared Hilbert–Schmidt norm in (14); multiplication by -2 proves the claim. Differences of couplings in $\Pi(\alpha, \beta)$ have zero marginals, so the last statement follows. □

Finite-dimensional coordinates. For finite spaces, let D, D' be the two distance matrices and let $\mathrm{J}_n = I_n - \mathbf{1}_n\mathbf{1}_n^\top/n$, with J_m defined similarly. The centered Gram matrices

$$\mathrm{G} = -\frac{1}{2}\mathrm{J}_n\mathrm{D}\mathrm{J}_n, \qquad \mathrm{G}' = -\frac{1}{2}\mathrm{J}_m\mathrm{D}'\mathrm{J}_m \tag{16}$$

are positive semidefinite. A zero-marginal signed matrix H satisfies

$$\mathcal{E}(\mathrm{H},\mathrm{H}) = -8\left\|\mathrm{G}^{1/2}\mathrm{H}\mathrm{G}'^{1/2}\right\|_{\mathrm{F}}^2. \tag{17}$$

This is the coordinate version of (14). It also shows when strictness fails: different couplings can be invisible to one of the centered geometries.

Corollary 3.2 (Tight squared-GW relaxation)**.** *Under the hypotheses of Proposition 3.1,*

$$\min_{\pi,\xi\in\Pi(\alpha,\beta)} \mathcal{E}(\pi,\xi) = \min_{\pi\in\Pi(\alpha,\beta)} \mathcal{E}(\pi,\pi). \tag{18}$$

Each component of a relaxed minimizer is a GW minimizer. If the tensor feature map in (14) *is injective on zero-marginal signed measures, every relaxed minimizer is diagonal.*

The conclusion is specific to the squared outer loss and to compatible signs of the two within-space kernels. For other distortions the polarized form remains linear in each coupling but may fail to be negative on feasible differences; the resulting two-plan problem is then only a lower bound.

4 Entropic and unregularized GW in one formulation

Choose a reference probability $\mathfrak{m} \in \mathcal{P}(\mathcal{Z})$ and use the lower-semicontinuous convention

$$\mathrm{KL}(\pi \mid \mathfrak{m}) = \begin{cases} \int (\rho\log\rho - \rho + 1)\,\mathrm{d}\mathfrak{m}, & \pi = \rho\mathfrak{m}, \\ +\infty, & \text{otherwise.} \end{cases} \tag{19}$$

For $\varepsilon \geq 0$, set

$$\mathcal{R}_\varepsilon(\pi) = \begin{cases} 0, & \varepsilon = 0, \\ \varepsilon\mathrm{KL}(\pi \mid \mathfrak{m}), & \varepsilon > 0, \end{cases} \tag{20}$$

so that no ambiguous product $0 \cdot (+\infty)$ occurs. Define

$$\mathcal{F}_\varepsilon(\pi) = \mathcal{E}(\pi,\pi) + \mathcal{R}_\varepsilon(\pi), \tag{21}$$

$$\mathcal{B}_\varepsilon(\pi,\xi) = \mathcal{E}(\pi,\xi) + \frac{1}{2}\mathcal{R}_\varepsilon(\pi) + \frac{1}{2}\mathcal{R}_\varepsilon(\xi). \tag{22}$$

The factors $1/2$ ensure $\mathcal{B}_\varepsilon(\pi,\pi) = \mathcal{F}_\varepsilon(\pi)$.

Corollary 4.1 (Tightness with and without entropy)**.** *Under* (3)*,*

$$\min_{\pi,\xi\in\mathcal{K}} \mathcal{B}_\varepsilon(\pi,\xi) = \min_{\pi\in\mathcal{K}} \mathcal{F}_\varepsilon(\pi) \qquad (\varepsilon \geq 0). \tag{23}$$

For $\varepsilon > 0$, every finite-valued relaxed minimizer is diagonal. For $\varepsilon = 0$, its two components are diagonal minimizers but need not coincide.

For GW, the canonical choice is

$$\mathfrak{m} = \alpha \otimes \beta, \qquad \mathcal{K} = \Pi(\alpha,\beta). \tag{24}$$

Then $\varepsilon = 0$ gives ordinary GW and $\varepsilon > 0$ gives entropic GW. In the discrete setting, $\mathrm{KL}(\pi \mid \alpha \otimes \beta)$ differs from the usual negative-entropy penalty on the coupling matrix only by a constant determined by the prescribed marginals.

5 Alternating minimization

The lifted problem suggests a one-sequence algorithm. Starting from $\pi^{(0)} \in \mathcal{K}$, define

$$\pi^{(\ell+1)} \in \arg\min_{\pi\in\mathcal{K}} \left\{ \mathcal{E}(\pi^{(\ell)}, \pi) + \frac{1}{2}\mathcal{R}_\varepsilon(\pi) \right\}. \tag{25}$$

This is exact alternating minimization of $\mathcal{B}_\varepsilon$, with the names of the two symmetric blocks exchanged after each update. It is also a majorization–minimization step for $\mathcal{F}_\varepsilon$.

Specialization to GW. For $\mathcal{K} = \Pi(\alpha, \beta)$, the frozen cost is

$$\mathcal{C}_{\pi^{(\ell)}}(x, y) = \int \left(d_\mathcal{X}(x, x') - d_\mathcal{Y}(y, y')\right)^2 \mathrm{d}\pi^{(\ell)}(x', y'). \tag{26}$$

Thus (25) is an ordinary OT problem with this cost. It is a linear Kantorovich problem when $\varepsilon = 0$ and an entropic OT problem, solvable by Sinkhorn, when $\varepsilon > 0$. The regularization strength of this block problem is $\varepsilon/2$, not ε, because each of the two lifted variables carries half of the diagonal entropy. In matrix coordinates the cost is

$$\mathrm{C}(\mathrm{P}) = \mathrm{D}^{\odot 2}\mathrm{a}\mathbf{1}_m^\top + \mathbf{1}_n(\mathrm{D}'^{\odot 2}\mathrm{b})^\top - 2\mathrm{D}\mathrm{P}\mathrm{D}'. \tag{27}$$

Proposition 5.1 (Descent of the diagonal objective)**.** *Assume* (3)*. Every exact update* (25) *satisfies*

$$\mathcal{F}_\varepsilon(\pi^{(\ell+1)}) \leq \mathcal{F}_\varepsilon(\pi^{(\ell)}). \tag{28}$$

This holds for every $\varepsilon \geq 0$, with any selected block minimizer when $\varepsilon = 0$.

Proof. Put $\pi = \pi^{(\ell)}$ and $\xi = \pi^{(\ell+1)}$. The tangent bound (12) gives $\mathcal{E}(\xi, \xi) \leq 2\mathcal{E}(\pi, \xi) - \mathcal{E}(\pi, \pi)$. Optimality of the block update, after multiplying by two, gives $2\mathcal{E}(\pi, \xi) + \mathcal{R}_\varepsilon(\xi) \leq 2\mathcal{E}(\pi, \pi) + \mathcal{R}_\varepsilon(\pi)$. Combining the inequalities proves (28). □

Fixed points and stationarity. Let $N_\mathcal{K}(\pi)$ denote the convex normal cone and interpret subdifferentials in the ambient signed-measure space. If π_* is a fixed point of (25), its block optimality condition, multiplied by two, is

$$0 \in 2\mathcal{C}_{\pi_*} + \partial\mathcal{R}_\varepsilon(\pi_*) + N_\mathcal{K}(\pi_*). \tag{29}$$

This is exactly first-order stationarity for $\mathcal{F}_\varepsilon$. In the entropic GW case, writing $\rho_* = \mathrm{d}\pi_*/\,\mathrm{d}(\alpha \otimes \beta)$, it takes the variational form

$$\int \left(2\mathcal{C}_{\pi_*} + \varepsilon \log \rho_*\right) \mathrm{d}(\pi - \pi_*) \geq 0 \qquad \text{for every } \pi \in \Pi(\alpha, \beta). \tag{30}$$

Equivalently, under the usual positivity assumptions, the expression in parentheses equals a sum of one potential on $\mathcal{X}$ and one potential on $\mathcal{Y}$, almost everywhere.

6 Intrinsic entropy descent on coupling measures

The useful geometry is already present in the regularizer. We therefore measure one block step by its entropy Bregman divergence rather than by a coordinate norm. Specialize here to $\mathcal{K} = \Pi(\alpha, \beta)$, put $\mathfrak{m} = \alpha \otimes \beta$, and identify $\pi = \rho\mathfrak{m}$ with its density. The feasible densities and their linear tangent space are

$$\mathcal{A} = \left\{ \rho \geq 0 : \int \rho(x,y)\,\mathrm{d}\beta(y) = 1,\ \int \rho(x,y)\,\mathrm{d}\alpha(x) = 1 \right\}, \tag{31}$$

$$\mathsf{T} = \left\{ h \in L^2(\mathfrak{m}) : \int h(x,y)\,\mathrm{d}\beta(y) = 0,\ \int h(x,y)\,\mathrm{d}\alpha(x) = 0 \right\}. \tag{32}$$

All equalities are understood almost everywhere. Let K be the integral operator induced by L,

$$(K\rho)(z) = \int_{\mathcal{Z}} L(z,z')\rho(z')\,\mathrm{d}\mathfrak{m}(z'), \qquad \mathcal{E}(\rho\mathfrak{m}, \eta\mathfrak{m}) = \langle \rho, K\eta \rangle_{L^2(\mathfrak{m})}. \tag{33}$$

Writing $\mathrm{Ent}(\rho) = \int(\rho\log\rho - \rho + 1)\,\mathrm{d}\mathfrak{m}$, the diagonal functional becomes

$$\Phi_\varepsilon(\rho) = \langle \rho, K\rho \rangle_{L^2(\mathfrak{m})} + \varepsilon \mathrm{Ent}(\rho). \tag{34}$$

6.1 An exact Bregman descent identity

The sign of the polarized GW form does more than prove monotonicity: it separates the decrease into two nonnegative terms.

Proposition 6.1 (Exact entropy descent)**.** *Let $\varepsilon > 0$, let $\rho^{(\ell)}, \rho^{(\ell+1)} \in \mathcal{A}$ be positive densities, and suppose that $\rho^{(\ell+1)}$ is the block update (25). Then*

$$\begin{aligned}\Phi_\varepsilon(\rho^{(\ell)}) - \Phi_\varepsilon(\rho^{(\ell+1)}) &= \varepsilon \mathrm{KL}\Big(\rho^{(\ell)}\mathfrak{m} \,\Big|\, \rho^{(\ell+1)}\mathfrak{m}\Big) \\ &\quad - \mathcal{E}\Big((\rho^{(\ell+1)} - \rho^{(\ell)})\mathfrak{m}, (\rho^{(\ell+1)} - \rho^{(\ell)})\mathfrak{m}\Big).\end{aligned} \tag{35}$$

Both terms on the right are nonnegative.

Proof. Put $\rho = \rho^{(\ell)}$, $\widehat{\rho} = \rho^{(\ell+1)}$, and $d = \widehat{\rho} - \rho \in \mathsf{T}$. Since the constraints in (31) are affine and the entropic block minimizer is positive, its first-order condition in the feasible direction $\rho - \widehat{\rho}$ is an equality:

$$\left\langle K\rho + \frac{\varepsilon}{2}\log\widehat{\rho}, \rho - \widehat{\rho} \right\rangle_{L^2(\mathfrak{m})} = 0.$$

The Bregman identity for negative entropy therefore gives

$$\langle \rho, K\rho \rangle + \frac{\varepsilon}{2}\mathrm{Ent}(\rho) - \langle \rho, K\widehat{\rho} \rangle - \frac{\varepsilon}{2}\mathrm{Ent}(\widehat{\rho}) = \frac{\varepsilon}{2}\mathrm{KL}(\rho\mathfrak{m} \mid \widehat{\rho}\mathfrak{m}).$$

The polarization identity now reads

$$\Phi_\varepsilon(\rho) - \Phi_\varepsilon(\widehat{\rho}) = 2\left[\langle \rho, K\rho \rangle + \frac{\varepsilon}{2}\mathrm{Ent}(\rho) - \langle \rho, K\widehat{\rho} \rangle - \frac{\varepsilon}{2}\mathrm{Ent}(\widehat{\rho})\right] - \langle d, Kd \rangle.$$

Substituting the preceding equality proves (35). Finally, $-\langle d, Kd \rangle = -\mathcal{E}(d\mathfrak{m}, d\mathfrak{m}) \geq 0$ because $d\mathfrak{m}$ is a feasible difference. $\square$

Since L is bounded and entropy is nonnegative, the objective is bounded below. Thus its values converge and

$$\sum_{\ell \geq 0} \mathrm{KL}\Big(\rho^{(\ell)}\mathfrak{m} \,\Big|\, \rho^{(\ell+1)}\mathfrak{m}\Big) < +\infty. \tag{36}$$

This conclusion is intrinsic to entropy. No total-variation estimate is needed.

6.2 A compact positive invariant set

For a bounded cost kernel, every entropic block problem is a Schrödinger scaling problem with a uniformly positive Gibbs kernel. This prevents the iterates from approaching the boundary of the positive cone.

Proposition 6.2 (Uniform positivity and compactness of GW updates)**.** *Assume that $\mathcal{X}, \mathcal{Y}$ are compact, that α, β have full support, and that L is continuous. Set $M = \|L\|_\infty$ and*

$$k_- = \exp(-2M/\varepsilon), \qquad k_+ = \exp(2M/\varepsilon), \qquad c_\varepsilon = (k_-/k_+)^2, \qquad C_\varepsilon = (k_+/k_-)^2.$$

Every block update has a continuous density satisfying

$$0 < c_\varepsilon \le \rho^{(\ell)}(x,y) \le C_\varepsilon < +\infty \qquad (\ell \ge 1). \tag{37}$$

The family $(\rho^{(\ell)})_{\ell \ge 1}$ is equicontinuous and hence relatively compact in $C(\mathcal{X} \times \mathcal{Y})$.

Proof. Since $\pi^{(\ell)}$ is a probability measure, $\|\mathcal{C}_{\pi^{(\ell)}}\|_\infty \le M$. The block Gibbs kernel

$$k_\ell(x,y) = \exp\Big(-\frac{2\mathcal{C}_{\pi^{(\ell)}}(x,y)}{\varepsilon}\Big)$$

therefore lies between k_- and k_+. The unique block minimizer has the scaling form $\rho^{(\ell+1)}(x,y) = u_\ell(x) k_\ell(x,y) v_\ell(y)$. Use the scalar indeterminacy of the two scalings to impose $\int v_\ell \,\mathrm{d}\beta = 1$. The two marginal equations then give

$$k_+^{-1} \le u_\ell \le k_-^{-1}, \qquad k_-/k_+ \le v_\ell \le k_+/k_-,$$

which proves (37).

Uniform continuity of L gives a common modulus of continuity for all frozen costs, hence for all k_ℓ. The formulas

$$u_\ell(x) = \Big(\int k_\ell(x,y) v_\ell(y) \,\mathrm{d}\beta(y)\Big)^{-1}, \qquad v_\ell(y) = \Big(\int k_\ell(x,y) u_\ell(x) \,\mathrm{d}\alpha(x)\Big)^{-1}$$

and the preceding lower bounds transfer this modulus to the scalings and to $\rho^{(\ell+1)}$. Arzelà–Ascoli gives relative compactness. □

On the box (37), Taylor's formula for $s \mapsto s \log s - s + 1$ yields

$$\frac{1}{2C_\varepsilon} \|\rho - \eta\|^2_{L^2(\mathfrak{m})} \le \mathrm{KL}(\rho \mathfrak{m} \mid \eta \mathfrak{m}) \le \frac{1}{2c_\varepsilon} \|\rho - \eta\|^2_{L^2(\mathfrak{m})}. \tag{38}$$

Consequently, (36) implies square-summability of the density increments in $L^2(\mathfrak{m})$.

Corollary 6.3 (Stationarity of cluster points)**.** *Under the assumptions of Proposition 6.2, the objective values converge, the density increments tend to zero in $L^2(\mathfrak{m})$, and every uniform cluster point is a fixed point of the block map and satisfies (30).*

Proof. Only the last claim remains to prove. Let $\rho^{(\ell_k)} \to \rho_*$ uniformly. Relative compactness gives a uniformly convergent subsequence of $\rho^{(\ell_k+1)}$; its limit equals ρ_*, because the corresponding L^2 increments tend to zero and $\mathfrak{m}$ has full support. For every $\eta \in \mathcal{A}$ of finite entropy, block optimality gives

$$\Big\langle \rho^{(\ell_k)}, K\rho^{(\ell_k+1)} \Big\rangle + \frac{\varepsilon}{2}\mathrm{Ent}(\rho^{(\ell_k+1)}) \le \Big\langle \rho^{(\ell_k)}, K\eta \Big\rangle + \frac{\varepsilon}{2}\mathrm{Ent}(\eta).$$

For infinite-entropy competitors the limiting inequality is automatic. The uniform density bounds make entropy continuous along this subsequence, while boundedness of K gives continuity of the bilinear terms. Passing to the limit shows that ρ_* uniquely minimizes the block functional frozen at ρ_*. Thus it is a fixed point. Since it is uniformly positive, its affine first-order condition is

$$\int (2K\rho_* + \varepsilon \log \rho_*) h \, \mathrm{d}\mathfrak{m} = 0 \qquad (h \in \mathsf{T}),$$

which is equivalent to (30). □

This still does not by itself prove convergence of the full sequence: square-summable increments need not have summable lengths. The next section supplies the missing finite-length mechanism.

7 Full convergence of alternating minimization

The final step is a measure-space analogue of the Kurdyka–Łojasiewicz argument used for finite-dimensional block minimization (Attouch et al., 2010; Xu and Yin, 2013). The correct local quadratic structure is the Fisher metric generated by entropy, not a Frobenius norm attached to a discretization. At a positive density ρ, it is

$$\|h\|_{\rho,\mathrm{F}}^2 = \int \frac{h^2}{\rho} \, \mathrm{d}\mathfrak{m}, \qquad h \in \mathsf{T}. \tag{39}$$

The bounds (37) make this norm uniformly equivalent to $L^2(\mathfrak{m})$ along the iterates. The proof has three ingredients: the exact entropy decrease controls the squared step, block optimality controls the residual gradient by the preceding step, and an infinite-dimensional Łojasiewicz–Simon inequality turns these two quadratic estimates into summable lengths.

7.1 Projected gradient and relative error

Let P_T be the $L^2(\mathfrak{m})$-orthogonal projection onto T. It has the explicit bounded extension to $L^\infty(\mathfrak{m})$

$$\begin{aligned}(\mathsf{P}_\mathsf{T} q)(x,y) = q(x,y) - \int q(x,y') \, \mathrm{d}\beta(y') - \int q(x',y) \, \mathrm{d}\alpha(x') \\ + \iint q(x',y') \, \mathrm{d}\alpha(x') \, \mathrm{d}\beta(y').\end{aligned} \tag{40}$$

The constrained gradient of Φ_ε is represented by

$$G_\varepsilon(\rho) = \mathsf{P}_\mathsf{T}(2K\rho + \varepsilon \log \rho). \tag{41}$$

Proposition 7.1 (Block residual estimate)**.** *The entropic GW update satisfies*

$$\mathsf{P}_\mathsf{T}\left(K\rho^{(\ell)} + \frac{\varepsilon}{2} \log \rho^{(\ell+1)}\right) = 0, \tag{42}$$

$$G_\varepsilon(\rho^{(\ell+1)}) = 2\mathsf{P}_\mathsf{T} K(\rho^{(\ell+1)} - \rho^{(\ell)}). \tag{43}$$

In particular, for a constant b depending only on L,

$$\left\|G_\varepsilon(\rho^{(\ell+1)})\right\|_{L^\infty(\mathfrak{m})} \le b \left\|\rho^{(\ell+1)} - \rho^{(\ell)}\right\|_{L^2(\mathfrak{m})}. \tag{44}$$

Proof. Equation (42) is the first-order condition of the affine-constrained block problem. Subtracting twice this identity from (41) at $\rho^{(\ell+1)}$ gives (43). Finally, P_T is bounded on L^∞, while

$$\|Kh\|_\infty \leq \|L\|_\infty \|h\|_{L^1(\mathfrak{m})} \leq \|L\|_\infty \|h\|_{L^2(\mathfrak{m})} .$$

□

7.2 The finite-length principle

The following theorem isolates the convergence mechanism. Its two ambient spaces play different roles: compactness is required in a strong Banach topology, whereas the length is measured in a weaker Hilbert norm.

Theorem 7.2 (Measure-space Łojasiewicz–Simon convergence principle)**.** *Let* B *be a Banach space continuously and injectively embedded in a Hilbert space* H*, and let* G *be a normed space. Let* $(\rho^{(\ell)})_\ell$ *be relatively compact in* B*, let* Φ *be continuous on the* B*-closure of the iterates, and let* $\Phi(\rho^{(\ell)})$ *decrease to* Φ_**. Assume that a residual map* G *and constants* $a, b > 0$ *satisfy*

$$\Phi(\rho^{(\ell)}) - \Phi(\rho^{(\ell+1)}) \geq a \left\|\rho^{(\ell+1)} - \rho^{(\ell)}\right\|_\mathsf{H}^2 , \tag{45}$$

$$\left\|G(\rho^{(\ell+1)})\right\|_\mathsf{G} \leq b \left\|\rho^{(\ell+1)} - \rho^{(\ell)}\right\|_\mathsf{H} . \tag{46}$$

Suppose that every cluster point ρ_* *is stationary and satisfies a Łojasiewicz–Simon inequality: there are a* B*-neighborhood* U_**, a constant* $c_* > 0$*, and* $\theta_* \in (0, 1/2]$ *such that*

$$|\Phi(\rho) - \Phi(\rho_*)|^{1-\theta_*} \leq c_* \|G(\rho)\|_\mathsf{G} \qquad (\rho \in U_*). \tag{47}$$

Then the sequence has finite length,

$$\sum_{\ell=0}^{+\infty} \left\|\rho^{(\ell+1)} - \rho^{(\ell)}\right\|_\mathsf{H} < +\infty, \tag{48}$$

and converges in H *to one stationary point.*

Proof. Choose a cluster point ρ_*, with convergence along a subsequence in B. Continuity of Φ gives $\Phi(\rho_*) = \Phi_*$. On the compact B-closure $\mathfrak{K}$ of the iterates, the embedding into H is a continuous injection into a Hausdorff space. Its inverse on its image is therefore continuous. Consequently, there is $\delta > 0$ such that

$$\rho \in \mathfrak{K}, \quad \|\rho - \rho_*\|_\mathsf{H} < \delta \quad \Longrightarrow \quad \rho \in U_*. \tag{49}$$

Write

$$e_\ell = \Phi(\rho^{(\ell)}) - \Phi_*, \qquad d_\ell = \left\|\rho^{(\ell+1)} - \rho^{(\ell)}\right\|_\mathsf{H} , \qquad \varphi(s) = \frac{c_*}{\theta_*} s^{\theta_*}.$$

Whenever $\rho^{(\ell)} \in U_*$, $e_\ell > 0$, and $d_{\ell-1} > 0$, concavity of φ, (45), (47) at $\rho^{(\ell)}$, and (46) at the preceding step give

$$\varphi(e_\ell) - \varphi(e_{\ell+1}) \geq \varphi'(e_\ell)(e_\ell - e_{\ell+1}) \geq \frac{a\, d_\ell^2}{b\, d_{\ell-1}}.$$

If $d_{\ell-1} = 0$, the relative-error and Łojasiewicz–Simon inequalities imply $e_\ell = 0$. If $e_\ell = 0$, monotonicity and sufficient decrease give $e_{\ell+1} = d_\ell = 0$, and induction makes the remaining tail stationary. Otherwise, the arithmetic–geometric mean inequality gives

$$2d_\ell \leq d_{\ell-1} + \frac{b}{a}(\varphi(e_\ell) - \varphi(e_{\ell+1})).$$

Because $d_\ell \to 0$ by sufficient decrease and $e_\ell \to 0$, choose an index n on the cluster subsequence such that

$$\left\|\rho^{(n)} - \rho_*\right\|_{\mathsf{H}} + d_{n-1} + \frac{b}{a}\varphi(e_n) < \delta. \tag{50}$$

Suppose, by contradiction, that the tail first exits U_* at $\rho^{(J+1)}$. The preceding estimate is valid for $\ell = n, \ldots, J$. Summing it and cancelling common increments gives

$$\sum_{\ell=n}^{J} d_\ell + d_J \le d_{n-1} + \frac{b}{a}(\varphi(e_n) - \varphi(e_{J+1})) \le d_{n-1} + \frac{b}{a}\varphi(e_n). \tag{51}$$

Together with (50), this places $\rho^{(J+1)}$ within δ of ρ_* in H, so (49) implies $\rho^{(J+1)} \in U_*$, a contradiction. The tail is therefore trapped, and letting $J \to +\infty$ in (51) proves finite length. The sequence is Cauchy in H and converges to ρ_*, because the selected cluster subsequence converges to that same point in H. □

7.3 Verification for entropic GW

For compact GW, the required infinite-dimensional gradient inequality is not an extra numerical assumption. It follows from analyticity of entropy and compactness of the integral operator.

$$\mathsf{T}_\infty = \mathsf{T} \cap L^\infty(\mathfrak{m}), \qquad \|h\|_{\mathsf{T}_\infty} = \|h\|_{L^\infty(\mathfrak{m})}.$$

This is a closed Banach subspace because the two marginal operators are bounded on L^∞.

Lemma 7.3 (The constrained entropy Hessian is invertible)**.** *Let $\rho \in \mathcal{A} \cap L^\infty(\mathfrak{m})$ satisfy $0 < c \le \rho \le C < +\infty$. Then*

$$\mathsf{A}_\rho : \mathsf{T}_\infty \longrightarrow \mathsf{T}_\infty, \qquad \mathsf{A}_\rho h = \mathsf{P}_{\mathsf{T}}\left(\frac{h}{\rho}\right), \tag{52}$$

is a bounded isomorphism.

Proof. Boundedness is immediate. If $\mathsf{A}_\rho h = 0$, then (40) shows that $h/\rho = a(x) + b(y)$ for some bounded a, b. Since h has zero marginals,

$$\int \frac{h^2}{\rho}\,\mathrm{d}\mathfrak{m} = \int h(x,y)(a(x) + b(y))\,\mathrm{d}\mathfrak{m}(x,y) = 0,$$

and hence $h = 0$. Thus A_ρ is injective.

To prove surjectivity, fix $g \in \mathsf{T}_\infty$. Introduce the two Markov operators

$$(Rb)(x) = \int \rho(x,y) b(y)\,\mathrm{d}\beta(y), \qquad (R^* a)(y) = \int \rho(x,y) a(x)\,\mathrm{d}\alpha(x),$$

and the bounded functions

$$r_{\mathcal{X}}(x) = \int \rho(x,y) g(x,y)\,\mathrm{d}\beta(y), \qquad r_{\mathcal{Y}}(y) = \int \rho(x,y) g(x,y)\,\mathrm{d}\alpha(x).$$

We seek $h = \rho(g + a + b)$. Its two marginals vanish precisely when

$$a + Rb = -r_{\mathcal{X}}, \qquad R^* a + b = -r_{\mathcal{Y}}. \tag{53}$$

Eliminating b gives

$$(I - RR^*)a = -r_{\mathcal{X}} + R r_{\mathcal{Y}}. \tag{54}$$

Denote the right-hand side by r. It has zero α-mean. The Markov operator $P = RR^*$ has transition density

$$q(x,x') = \int \rho(x,y)\rho(x',y)\,\mathrm{d}\beta(y) \geq c^2$$

with respect to α. Its rows integrate to one and, because $q \geq c^2$, it admits the minorization $q(x,\cdot)\alpha \geq c^2\alpha$. Hence, for every bounded a,

$$\operatorname{osc}(Pa) \leq (1-c^2)\operatorname{osc}(a), \qquad \operatorname{osc}(a) = \operatorname{ess\,sup} a - \operatorname{ess\,inf} a. \tag{55}$$

Indeed, subtracting the common component $c^2\alpha$ leaves a Markov kernel of mass $1-c^2$. Thus P is a strict contraction on $L^\infty(\alpha)/\mathbb{R}\mathbf{1}$, and the Neumann series $\sum_{n\geq 0} P^n r$ converges in that quotient. Since P preserves α and r has zero mean, its zero-mean representative is a bounded solution a of (54). Moreover, $\|r_{\mathcal{X}}\|_\infty, \|r_{\mathcal{Y}}\|_\infty \leq \|g\|_\infty$, so (55) bounds $\|a\|_\infty$ by a constant depending only on c times $\|g\|_\infty$.

Set $b = -r_{\mathcal{Y}} - R^*a$. Then (53) holds, so $h = \rho(g+a+b)$ lies in T_∞. Finally, $\mathsf{P}_\mathsf{T}(h/\rho) = g$, because the projection annihilates $a(x)+b(y)$, and $\|h\|_\infty \leq C_\rho \|g\|_\infty$. This proves surjectivity and boundedness of A_ρ^{-1}. □

Proposition 7.4 (Łojasiewicz–Simon inequality for entropic GW)**.** *Under the assumptions of Proposition 6.2, every strictly positive stationary density $\rho_* \in \mathcal{A} \cap C(\mathcal{X}\times\mathcal{Y})$ satisfies (47) with $\Phi = \Phi_\varepsilon$, $G = G_\varepsilon$, $\mathsf{B} = \mathsf{G} = \mathsf{T}_\infty$.*

Proof. Set $\mathsf{X} = \mathsf{T}_\infty$. The $L^2(\mathfrak{m})$ pairing defines a continuous embedding

$$\jmath : \mathsf{X} \longrightarrow \mathsf{X}^*, \qquad \jmath(q)[h] = \int hq\,\mathrm{d}\mathfrak{m}.$$

It is definite because $\jmath(q)[q] = \|q\|^2_{L^2(\mathfrak{m})} > 0$ for $q \neq 0$. On the open set

$$U_* = \{h \in \mathsf{X} : \operatorname{ess\,inf}(\rho_* + h) > 0\},$$

define

$$\mathcal{E}_*(h) = \Phi_\varepsilon(\rho_* + h), \qquad \mathcal{M}_*(h) = G_\varepsilon(\rho_* + h) \in \mathsf{X}.$$

Stationarity and strict positivity give $\mathcal{M}_*(0) = 0$. The quadratic interaction is analytic on L^∞, and the Nemytskii map $\rho \mapsto \log\rho$ is real analytic on every uniformly positive L^∞-ball. Hence $\mathcal{E}_*$ and $\mathcal{M}_*$ are real analytic. Moreover, for $h, k \in \mathsf{X}$, self-adjointness of P_T in $L^2(\mathfrak{m})$ and $k \in \mathsf{T}$ give

$$D\mathcal{E}_*(h)[k] = \int k\,\mathcal{M}_*(h)\,\mathrm{d}\mathfrak{m},$$

so $\mathcal{M}_*$ is a gradient map for $\mathcal{E}_*$ with values in $\mathsf{X} \subset \mathsf{X}^*$. Its derivative at the critical point $h = 0$ is

$$D\mathcal{M}_*(0)h = 2\mathsf{P}_\mathsf{T}Kh + \varepsilon\mathsf{A}_{\rho_*}h, \qquad h \in \mathsf{T}_\infty. \tag{56}$$

Because L is uniformly continuous on the compact product, the image by K of the L^∞-unit ball is uniformly bounded and equicontinuous in $C(\mathcal{X}\times\mathcal{Y})$. Arzelà–Ascoli and the boundedness of P_T on continuous functions show that $\mathsf{P}_\mathsf{T}K : \mathsf{X} \to \mathsf{X}$ is compact. By Lemma 7.3, (56) is an isomorphism plus a compact operator and is Fredholm of index zero. The analytic-gradient-map Łojasiewicz–Simon theorem of Feehan and Maridakis (2020) applies with the definite embeddings $\mathsf{X} = \widetilde{\mathsf{X}} = \mathsf{T}_\infty \hookrightarrow \mathsf{X}^*$. It yields constants $Z, \sigma > 0$ and $\vartheta \in [1/2, 1)$ such that

$$\|G_\varepsilon(\rho)\|_\infty \geq Z\,|\Phi_\varepsilon(\rho) - \Phi_\varepsilon(\rho_*)|^\vartheta \quad \text{for } \|\rho - \rho_*\|_\infty < \sigma,$$

for feasible ρ in the positive chart. Setting $\theta_* = 1 - \vartheta \in (0, 1/2]$ and $c_* = Z^{-1}$ gives (47). This is the precise infinite-dimensional replacement for the finite-dimensional KL property; see also Chill (2003). □

Theorem 7.5 (Full convergence of entropic alternating GW on measures)**.** *Assume that $\mathcal{X}, \mathcal{Y}$ are compact metric spaces, replace them by the supports of α, β, and assume that the symmetric GW kernel L is continuous and negative on feasible differences. For every fixed $\varepsilon > 0$, initialize (25) with a positive continuous coupling density, for instance $\rho^{(0)} = 1$. Then there is a continuous stationary fixed-point density ρ_*, satisfying $c_\varepsilon \leq \rho_* \leq C_\varepsilon$, such that*

$$\sum_{\ell=0}^{+\infty} \left\| \rho^{(\ell+1)} - \rho^{(\ell)} \right\|_{L^2(\mathfrak{m})} < +\infty, \qquad \rho^{(\ell)} \longrightarrow \rho_* \quad \text{uniformly.} \tag{57}$$

Equivalently, $\pi^{(\ell)} = \rho^{(\ell)}\mathfrak{m}$ converges to the stationary coupling $\pi_* = \rho_*\mathfrak{m}$. The theorem does not claim that π_* is a global GW minimizer, and its constants are not uniform as $\varepsilon \downarrow 0$.

Proof. Proposition 6.1 and (38) give (45) after the first, immaterial update; explicitly one may take $a = \varepsilon/(2C_\varepsilon)$. Proposition 7.1 gives (46). The iterates are relatively compact in the uniform topology by Proposition 6.2, Φ_ε is continuous on their uniformly positive closure, and every cluster point is stationary. Proposition 7.4 and Theorem 7.2 apply to the shifted iterates $h^{(\ell)} = \rho^{(\ell)} - 1$, the shifted functional $\widehat{\Phi}_\varepsilon(h) = \Phi_\varepsilon(1 + h)$, the residual $\widehat{G}_\varepsilon(h) = G_\varepsilon(1 + h)$, and

$$\mathsf{B} = \mathsf{T}_\infty, \qquad \mathsf{H} = \mathsf{T} \subset L^2(\mathfrak{m}), \qquad \mathsf{G} = \mathsf{T}_\infty,$$

They prove finite length and L^2-convergence to a stationary density ρ_*. To upgrade the convergence, suppose that uniform convergence failed. A subsequence would remain a fixed positive uniform distance from ρ_*; relative compactness in $C(\mathcal{X} \times \mathcal{Y})$ would give a uniformly convergent subsequence. Its limit must equal ρ_* in $L^2(\mathfrak{m})$, and full support of $\mathfrak{m}$ identifies the two continuous representatives, a contradiction. $\square$

7.4 Extension beyond quadratic GW

Only two algorithmic estimates were used: each exact block minimization must produce a quadratic Bregman decrease, and changing the other block must control the residual gradient. This gives a genuinely biconvex measure-space statement.

Theorem 7.6 (Analytic biconvex alternating minimization)**.** *Let $\mathsf{X} = x_0 + \mathsf{X}_0$ and $\mathsf{Y} = y_0 + \mathsf{Y}_0$ be affine spaces modeled on Banach spaces $\mathsf{X}_0, \mathsf{Y}_0$, continuously and injectively embedded in Hilbert spaces $\mathsf{H}_\mathsf{X}, \mathsf{H}_\mathsf{Y}$. Let $\mathcal{B}$ be analytic on a neighborhood of $\mathsf{X} \times \mathsf{Y}$ and convex in each block. Consider the exact updates*

$$y^{(\ell+1)} \in \arg\min_y \mathcal{B}(x^{(\ell)}, y), \qquad x^{(\ell+1)} \in \arg\min_x \mathcal{B}(x, y^{(\ell+1)}). \tag{58}$$

All derivatives below are restricted to the model spaces X_0 and Y_0, and all product spaces carry their standard Euclidean product norms. Assume that the iterates have compact closure in the product Banach topology, that $\mathcal{B}$ is continuous there, and that the following properties hold on an invariant neighborhood of this closure:

(i) *both block minimizers are unique;*

(ii) *for some $a > 0$, each block update decreases $\mathcal{B}$ by at least $a/2$ times the squared norm of that block increment in its Hilbert tangent space;*

(iii) *the constrained cross derivative is Lipschitz: for some $b > 0$,*

$$\|D_y\mathcal{B}(x, y) - D_y\mathcal{B}(\widetilde{x}, y)\|_{\mathsf{Y}_0^*} \leq b\, \|x - \widetilde{x}\|_{\mathsf{H}_\mathsf{X}};$$

(iv) *at every stationary cluster point* z_**, there are* $c_* > 0$*,* $\theta_* \in (0, 1/2]$*, and a product-Banach neighborhood on which*

$$|\mathcal{B}(z) - \mathcal{B}(z_*)|^{1-\theta_*} \leq c_* \|D\mathcal{B}(z)\|_{\mathsf{X}_0^* \times \mathsf{Y}_0^*}.$$

Then the full-cycle sequence $z^{(\ell)} = (x^{(\ell)}, y^{(\ell)})$ *has finite length in* $\mathsf{H}_\mathsf{X} \times \mathsf{H}_\mathsf{Y}$ *and converges to one stationary point of* $\mathcal{B}$*.*

Proof. Adding the two block-decrease inequalities gives

$$\mathcal{B}(z^{(\ell)}) - \mathcal{B}(z^{(\ell+1)}) \geq \frac{a}{2}\left(\left\|x^{(\ell+1)} - x^{(\ell)}\right\|_{\mathsf{H}_\mathsf{X}}^2 + \left\|y^{(\ell+1)} - y^{(\ell)}\right\|_{\mathsf{H}_\mathsf{Y}}^2\right). \tag{59}$$

Continuity on the compact closure bounds $\mathcal{B}$ below, so (59) makes both block increments tend to zero.

At the end of a cycle, $D_x\mathcal{B}(x^{(\ell+1)}, y^{(\ell+1)}) = 0$. Optimality of the preceding y-update gives $D_y\mathcal{B}(x^{(\ell)}, y^{(\ell+1)}) = 0$. Therefore assumption (iii) yields the relative-error estimate

$$\begin{aligned}\left\|D\mathcal{B}(z^{(\ell+1)})\right\|_{\mathsf{X}_0^* \times \mathsf{Y}_0^*} &= \left\|D_y\mathcal{B}(x^{(\ell+1)}, y^{(\ell+1)}) - D_y\mathcal{B}(x^{(\ell)}, y^{(\ell+1)})\right\|_{\mathsf{Y}_0^*} \\ &\leq b\left\|x^{(\ell+1)} - x^{(\ell)}\right\|_{\mathsf{H}_\mathsf{X}}.\end{aligned} \tag{60}$$

Thus the constrained gradient tends to zero. Compactness and continuity of $D\mathcal{B}$ make every cluster point stationary, so assumption (iv) applies at each of them. Equations (59) and (60) are precisely the sufficient-decrease and relative-error estimates of Theorem 7.2, with $\mathsf{B} = \mathsf{X}_0 \times \mathsf{Y}_0$, $\mathsf{H} = \mathsf{H}_\mathsf{X} \times \mathsf{H}_\mathsf{Y}$, and $\mathsf{G} = \mathsf{X}_0^* \times \mathsf{Y}_0^*$. That theorem gives finite length and convergence to one stationary point. □

For integral energies, assumption (ii) is naturally furnished by a uniformly coercive Legendre or entropy Hessian, while assumption (iii) follows from a bounded cross-kernel. Analyticity and a Fredholm constrained Hessian imply assumption (iv) by the same argument as Proposition 7.4. The theorem therefore covers bilinear and biquadratic interactions, and more general smooth biconvex integral functionals, over spaces of measures. It is the infinite-dimensional counterpart of classical block-coordinate convergence theory (Tseng, 2001).

Remark 7.7 (Why positive regularization matters). For $\varepsilon = 0$, a GW block update is a linear problem and can have many minimizers. The Bregman term in (35), the positive density box, and the coercive part of (56) all disappear. Monotonicity from Proposition 5.1 remains, but a selection rule can cycle on a constant-value face. Full convergence of the unregularized iteration therefore requires a separate uniqueness, proximal, or active-set hypothesis.

8 Practical conclusions

The measure formulation separates properties that are often conflated.

Question	Correct statement
What is duplicated?	A measure $\pi \in \mathcal{K}$: the diagonal form $\mathcal{E}(\pi, \pi)$ becomes $\mathcal{E}(\pi, \xi)$.
Why is the relaxation tight?	Because $\mathcal{E}$ is nonpositive on $\mathrm{span}(\mathcal{K} - \mathcal{K})$, as exposed by (11).
Why does this apply to GW?	Differences of couplings have zero marginals, and negative-type geometry gives the square identity (14).
What is one block update?	Linear OT for $\varepsilon = 0$, entropic OT with strength $\varepsilon/2$ for $\varepsilon > 0$.
What converges on compact spaces?	For entropic GW, the whole sequence converges uniformly to a stationary positive coupling density and has finite Fisher-equivalent L^2 length.
When does the whole sequence converge?	Whenever sufficient Bregman decrease, a relative-gradient error bound, precompactness, and a Łojasiewicz–Simon inequality hold; these properties are verified here for compact entropic GW.
Does it find the global GW minimum?	Not in general. The algorithm remains a local method for a nonconvex diagonal objective.

9 Conclusion

The basic GW iteration is an instance of a general measure-level construction. A negative symmetric bilinear form can be polarized without changing its global diagonal minimum, while each variable of the polarized problem enters linearly. Squared GW inherits the required negativity from the Hilbert embeddings of negative-type distances. Entropy fits as a separable strictly convex penalty and turns every frozen-measure problem into a Sinkhorn solve. This common structure explains tightness, the factor $\varepsilon/2$, descent, and stationarity in one argument. More importantly, it exposes the right convergence geometry. The exact decrease is a relative entropy, its local quadratic form is the Fisher metric, and the linearized constrained gradient is an entropy isomorphism perturbed by a compact GW operator. The resulting Łojasiewicz–Simon argument proves finite length and convergence directly for measure-valued iterates, without passing through a matrix norm. The same proof architecture extends to analytic biconvex integral energies with coercive block geometry and controlled cross derivatives.

References

Hedy Attouch, Jérôme Bolte, Patrick Redont, and Antoine Soubeyran. Proximal alternating minimization and projection methods for nonconvex problems: An approach based on the Kurdyka–Łojasiewicz inequality. *Mathematics of Operations Research*, 35(2):438–457, 2010. doi: 10.1287/moor.1100.0449.

Ralph Chill. On the łojasiewicz–simon gradient inequality. *Journal of Functional Analysis*, 201(2): 572–601, 2003. doi: 10.1016/S0022-1236(02)00102-7.

Paul M. N. Feehan and Manousos Maridakis. Łojasiewicz–simon gradient inequalities for analytic and Morse–Bott functions on Banach spaces. *Journal für die reine und angewandte Mathematik*, 2020(765):35–67, 2020. doi: 10.1515/crelle-2019-0029.

Hiroshi Konno. Maximization of a convex quadratic function under linear constraints. *Mathematical Programming*, 11(1):117–127, 1976. doi: 10.1007/BF01580380.

Isaac J. Schoenberg. Metric spaces and positive definite functions. *Transactions of the American Mathematical Society*, 44(3):522–536, 1938. doi: 10.2307/1989894.

Thibault Séjourné, François-Xavier Vialard, and Gabriel Peyré. The unbalanced gromov–wasserstein distance: Conic formulation and relaxation. In *Advances in Neural Information Processing Systems*, volume 34, 2021. URL https://arxiv.org/abs/2009.04266.

Paul Tseng. Convergence of a block coordinate descent method for nondifferentiable minimization. *Journal of Optimization Theory and Applications*, 109(3):475–494, 2001. doi: 10.1023/A:1017501703105.

Yangyang Xu and Wotao Yin. A block coordinate descent method for regularized multiconvex optimization with applications to nonnegative tensor factorization and completion. *SIAM Journal on Imaging Sciences*, 6(3):1758–1789, 2013. doi: 10.1137/120887795.